Evaluating Bidirectional Label-Flipping and a Hybrid KNN-OOF Defense for ML-Based Network Intrusion Detection

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Abstract— **Machine learning-based network intrusion detection systems play a significant role in detecting malicious network traffic. Moreover, they are primarily responsible for protecting computers from cyberattacks. Yet the efficiency of these systems is directed by the quality and integrity of the data, leaving them vulnerable to many types of data poisoning attacks. One such technique, label-flipping, alters the training labels without changing the characteristics of the initial traffic. This method compromises the effectiveness of the intrusion detection system. In this study, a hybrid K-Nearest Neighbors and Out-of-Fold (KNN-OOF) defense mechanism is proposed, and the influence of label-flipping attacks on ML-based intrusion detection systems is studied in both directions. The research involved the training of classifiers, namely Random Forest and Decision Tree, using the UNSW-NB15 dataset and testing using clean data and poisoned data involving 10%, 20%, and 30% poisoning rates. The performance evaluation is carried out by measuring metrics including accuracy, precision, recall, F1 score, false negative rate, and false positive rate to understand the strengths and weaknesses of the defense mechanism against bidirectional label-flipping attacks.**

Keywords— **Network Intrusion Detection Systems (NIDS), Adversarial Machine Learning, Data Poisoning, Label-Flipping Attack, UNSW-NB15, KNN, Out-of-Fold Prediction**

I. INTRODUCTION

A. Problem Statement

Nowadays, many networks are subject to cyberattacks in which attackers gain access and interfere with the network maliciously. Network intrusion is a well-known cybersecurity threat, and with ongoing technological advances, many machine-learning-based models have been increasingly used and extensively trained to distinguish between normal and malicious traffic. However, the performance of such models depends heavily on the quality of the data they are trained on, making them vulnerable to data poisoning and cyberattacks. One such attack is label flipping, in which an attacker manipulates the training data by switching the labels of some samples, such as labeling attack traffic as benign or benign traffic as attack. This can cause the model to learn incorrect classifications and patterns, affecting its performance by reducing its ability to identify malicious traffic accurately.

B. Motivation

These reduced detection capabilities can have serious consequences for individuals and organizations, as network intrusion attacks may go undetected. Through these label-flipping attacks, other cyberattacks can bypass the system, allowing for unauthorized access to the system, stealing sensitive information, disrupting services, and causing data breaches. Because of this, ML-based Network Intrusion Detectors need to be accurate and reliable so that organizations and individuals can use them efficiently without letting malicious intrusion attempts bypass detection. Therefore, analyzing the impact of label-flipping attacks on ML-based NIDS is crucial for understanding how vulnerable they are to these threats and how their performance and capabilities can be affected. This can further help us identify ways to make them more resistant to such attacks, improving robustness and reliability.

C. Scope

This study examines the effects of bidirectional label-flipping attacks on centralized ML-based Network Intrusion Detection Systems (NIDS). The system will be trained using the publicly available

UNSW-NB15 dataset for binary classification, where each network traffic sample is classified as either Attack (1) or Benign (0). Random bidirectional label-flipping attacks will be simulated at different poisoning rates, including 10%, 20%, and 30% of the training set, to assess their impact on the effectiveness of the models in distinguishing between benign and malicious traffic. The study will focus on Random Forest and Decision Tree classifiers and will evaluate their performance on the clean dataset, poisoned dataset, and after applying the proposed defense mechanism. Additionally, metrics such as accuracy, precision, recall, F1 score, false-negative rate, and false-positive rate will be used for evaluation. It focuses only on centralized machine learning while excluding federated and distributed learning approaches from the project's scope.

D. Relevance to Real-World Cybersecurity Practice

Network Intrusion Detection Systems (NIDS) are vital tools for identifying network attacks conveyed by malicious network traffic. ML-based NIDS can learn patterns from the previously labelled network traffic in the training dataset and use them to distinguish between benign and malicious traffic [1]. This indicates that training data plays a vital role and needs to be kept intact and safe for reliable intrusion detection. Attackers have gotten more advanced in this day and age, leveraging newly developed technologies like deep learning and machine learning to carry out malicious activities. Data poisoning attacks such as label-flipping could affect the system's ability to detect future attacks, leading to undetected attacks or increased false alarms [1].

Studying the effects of label-flipping attacks can therefore help cybersecurity professionals better understand the system's vulnerabilities. Similarly, evaluating the proposed defense mechanisms against these attacks may also help in developing more reliable NIDS that can maintain good performance and remain robust even against such attacks where part of the system is manipulated.

II. Literature Review

Table I summarizes the related works and papers reviewed in this study. The most frequently used datasets in the reviewed papers were UNSW-NB15, CICIDS2017, CSE-CICIDS2018, and NSL-KDD [2]-[13], which were used to evaluate models in the NIDS domain. These studies will help investigate and analyze the effects of label-flipping attacks and the effectiveness of the proposed defense mechanisms.

A. Attacks

Several papers investigate the effects of label-flipping attacks on machine learning models. This includes Dunn et al. [2], who tested different label-flipping poisoning ratios ranging from 0% to 30%. Their results showed that as the poisoning rates increased, model performance decreased across several classifiers, including Random Forest, Naïve Bayes, and others, with precision reaching as low as 47.1%.

Other papers, such as Paracha et al. [8], showed that the Random Forest model was more resilient, experiencing only an 8.5% decrease in accuracy, while Decision Tree had a 14.85% increase in false-positive rate at a 15% poisoning rate in UNSW-NB15. KNN also experienced a 15.48% decrease in accuracy at a 25% poisoning rate on BotDroid. These findings demonstrate that different models are affected differently by poisoned labels, showing their vulnerabilities. The decision tree was affected because corrupted labels influenced its splitting decisions, while KNN was affected by corrupted density classes. Mohammadian et al. [5] also showed that the accuracy of a Deep Neural Network (DNN) decreased from 97% to 29% at a 70% label-flipping poisoning rate, supporting the conclusion that label-flipping attacks can significantly degrade the performance of machine learning models.

Label-flipping attacks have also been investigated in federated learning settings, where models are trained across multiple clients and later aggregated together. Nowroozi et al. [7] demonstrated that label-flipping attacks caused a significant drop in accuracy, making the network server raise a critical error and easily detect it. Similarly, Lavaur et al. [10] showed that model accuracy could decrease to approximately 40-50%,

demonstrating that federated learning systems can also experience critical performance degradation.

B. Denfenses

The study by Lai et al. [6] proposes a two-phase defense mechanism against label-flipping attacks in federated learning-based intrusion detection. The first phase identifies the poisoned client models by comparing their model weights. The second phase further identifies the poisoned models based on the performance of the aggregated global model. Results of this defense system show that the proposed defense improved performance by 3 % compared with other existing defense mechanisms.

Other studies, such as Sameera et al. [12], proposed a defense called WeiDetect, which detects label-flipping attacks by evaluating local models using validation sets, then putting the results into a probability for analyzing outliers, making it exclude the low-performance models that were attacked. Their results showed improvements of up to 70% in recall and between 1% and 14% in F1-score compared with existing defenses, showing its efficiency.

Another paper [3] proposes SecFedNIDS, which detects poisoned models based on model parameters and uses unsupervised poisoned model detection to reject malicious models. It also identifies poisoned data through class-path similarity, improving accuracy by 13%. Luong et al. [14] proposed FedLSAE, which uses an autoencoder that compresses the features of the training data and checks for similarities to group them. The method was tested on two IoT cyberattack datasets using several ML models. Other studies, such as Rezai et al. [9], proposed GRAF-IDS, which uses an enhanced version of the FedAVG method, making aggregation through graph-based clustering, resulting in lower false positive rates.

For centralized learning, Jang et al. [13] use KNN sanitization together with a self-supervised learning model, BYOL, as a defense against label-flipping attacks. The BYOL model identifies useful feature representations, which are then analyzed using KNN to determine whether a sample's label disagrees with the labels of its nearest neighbors. Suspicious samples are then removed before model training. Their results showed that there was an improvement with a 0.779 F1-score in random label flipping attacks and 0.701 accuracy in decision boundary label flipping attacks.

Of particular relevance, He et al. [11] proposed TE-PADN, which uses temporal margin sample generation with an attention mechanism that generates these samples to fix poisoned models. They start by collecting the infected samples, then putting them into a model that has an attention mechanism which generates synthetic samples. These samples represent data points that are close to the true decision boundary. Then they fix the decision boundary that had shifted during the label-flipping attack by introducing these synthetic samples into the training dataset, which pulls the decision boundary back to its correct position. Their results showed an increase in accuracy by 6.5%, proving their defense mechanism effective.

C. Research Gaps

Several of the reviewed studies mainly investigate the impact of label-flipping attacks on ML-based models without implementing defense mechanisms. These studies demonstrate how poisoning attacks affect different classifiers but do not focus on methods for identifying or correcting poisoned labels. Therefore, investigating a defense mechanism is important to mitigate the problem with the help of the defense papers used [2], [5], [7], [8], [10].

Other reviewed studies proposed defense mechanisms, specifically in federated learning settings, using techniques such as model aggregation, clustering, validation-based detection, and poisoned client rejection. However, these techniques are mainly designed for federated environments and might not directly address poisoned training samples in a centralized setting [3], [4], [6], [9], [12], [14].

In centralized learning, KNN was used as a technique to identify suspicious samples based on neighborhood label disagreement. However, relying on this approach alone might not provide sufficient evidence that a label was poisoned. Additionally, some reviewed approaches remove suspicious samples rather than correcting them. Removing a large number of samples, specifically under higher

poisoning rates, will reduce the amount of training data available for the model, possibly affecting its performance [13].

Therefore, this introduces an opportunity to study a hybrid defense mechanism that combines neighborhood-based and model-based findings. Combining both approaches might provide us with stronger evidence for identifying suspicious labels and analyzing their impact on classification performance.

TABLE I
LITERATURE REVIEW PAPERS

Reference	Year	Objectives	Dataset	Methods	Results	Limitations
[2]	2020	Testing the robustness of machine learning models against data poisoning. Training against label manipulation in IoT.	UNSW-NB15, ToN_IoT	Random Forest Classifier, Gradient Boosting machine, Naïve Bayes, Feed-forward Deep Learning	Model performances degrade when number of normal training observations is not larger than manipulated data.	Test models with data poisoning defenses to determine their efficiency.
[3]	2022	Increasing robustness against data poisoning attacks using SecFedNIDS detecting poisoned models and poisoned data.	UNSW-NB15, CICIDS2018	Convolutional Neural Network, SecFedNIDS	Improved accuracy by 13%	No explicit limitation mentioned.
[4]	2023	To reduce the impact of poisoning attacks on federated learning-based intrusion detection in IoT networks by proposing a lightweight detection mechanism on the central server.	CICIDS2017	Loss-based scoring mechanism (LSM), K-means clustering, Manhattan similarity	The strategy improves the accuracy of the FL-based intrusion detection model on the CIC-IDS-2017 dataset from 84.3% to 97.1%, boosting IoT network security.	One of the study's limitations is the well-thought-out model of poisoning attacks. The number of rounds, the number of malicious clients, and the attack launch time affect its efficacy.
[5]	2023	Testing deep learning models' performance against various percentages of label flipping attacks from 10% to 70% on two benchmark datasets. Using different class ratios to show how dataset imbalance affects performance.	CIC-IDS2017, CSECIC-IDS2018	Deep Neural Network	Models' performance always drops at label flipping attacks where accuracy drops from 97% to 29% when 70% of labels are flipped.	Using only one model family. Need to find mitigations to the label flipping attacks. Using random sampling to create label-flipping attacks.
[6]	2023	Using two-phase defense mechanism to prevent label-flipping attacks in federated learning IDS by	CICIDS2017	CNN (Convolutional Neural Networks)	DPA-FL improving F1-score by 3% than other defense mechanisms and	Using IID dataset, using random sampling in AP.

		comparing weights in first phase and finding the attacker in the second phase.			can work on both random and PGD label-flipping attacks.	
[7]	2024	The severity of data poisoning on network servers. To explore the effects of label-flipping attacks as they are easy to implement, but difficult to detect.	UNSW-NB15, CICIDS2017	Random Forest Classifier	Label flipping attacks drop accuracy significantly making them easier to identify, while servers have a hard time identifying feature poisoning.	Experimenting data poisoning on more than 2 clients.
[14]	2024	Investigating three common types of poisoning attacks including data and model poisoning. The authors introduce a new anti-poisoning mechanism (Fed-LSAE) that use an autoencoder- based latent space representation.	CIC-ToN-IoT, N-BaloT	Using Latent Space Representation through penultimate layer and autoencoder.	The result is categorized into Benign and attack group and the benign is sent to the central server, and the attack one is being eliminated.	The design of this study targeted IoT devices, which have limited computational power, network constraints and diverse devices. Scalability issue, as the number of clients grows, keeping accurate clustering become more challenging.
[8]	2025	Analyzing effects of data poisoning on different machine learning models. Studying the correlation between data poisoning rates, varying from 5-25%, and the classification accuracy of the machine learning algorithms.	UNSW-NB15, BotDroid, CTU13, CICIDS2017	Decision Tree, Random Forest, KNN (k-nearest neighbors) , SVM (Support Vector Machine), Gaussian Naïve Bayes, Perceptron	Understanding model mechanisms with Random Forest being most resilient. Decision Tree and KNN were most vulnerable as the features split in decision tree were disrupted and KNN had corrupted boundaries of density classes.	Limited to binary classifiers and baseline ML models. Overfitting in SVM and Perceptron models unexplained.
[9]	2025	The work aims to introduce a strong and secure aggregation method for the Federated learning in IoT environment compared with existing aggregation methods such as FedAvg that may perform poorly against the advanced adversarial strategies as noise-based flipping attacks.	N-BaloT2018, UNSW-NB15	Graph-based clustering aggregation (GBCA)	After applying label-flipping attack (LFA) and noise-based label flipping attack (NLFA) with different percentages: 25%, 30% and 35% using different aggregation methods (Krum and FedAvg), GBCA get higher scores in F1 and Precision. And a lower score in False Positive Rate.	High time complexity of O(n^2 log n). Use more efficient search and clustering techniques. Use the algorithm with a wider range of adversarial attacks and tested under different network latency conditions.

[10]	2025	To analyze the impact of label flipping attacks on federated-learning based intrusion detection systems (FIDS)	CSE-CICIDS2018, UNSW-NB15	Multilayer Perceptron (MLP), FedAvg, Principal Component Analysis (PCA)	In trained centralized learning, MLP achieved 0.992 accuracy on CSE-CICIDS2018 and 0.995 accuracy on UNSW-NB15. In federated learning accuracy 0.95 accuracy without attacks, but with 50% label flipping it decreased to 0.4 to 0.5.	The sample size of 10 individuals for its investigations. Heterogeneity simulation relies on a simplified depiction that is not entirely realistic of a real-world context. The results of two datasets are not generalizable.
[11]	2025	Using temporal margin sample generation, which maps original data into a latent feature space learning the temporal characteristics of the data, then the attention mechanism generates temporal margin samples to repair poisoned models. Repairs models' classification boundaries that were shifted due to the label flipping.	NSL-KDD, UNSW-NB15, CICIDS2017	Deep Neural Network (DNN), Multi-layer Perceptron, Recurrent Neural Networks (RNN)	Accuracy increased by 6.5% on NSL-KDD, 5.3% in UNSW-NB15, and 5.9% in CICIDS2017 with their proposed defense.	Enhancing the applicability of their system across more diverse intrusion detection systems.
[12]	2025	WeiDetect a two-phase defense system in federated learning with NIDS. They evaluate models using validation dataset and detecting adversarial attacks.	CIC-Darknet2020, CSE-CIC-IDS2018	Krum, MKrum, TMean, Median, FLAME, WeiDetect	Achieved higher recall by 70%, and 1% to 14% better global F1-score.	Extending the proposed approach to be able to support streaming datasets, and asynchronous FL.
[13]	2026	Using self-supervised learning and KNN sanitization which involves KNN label disagreement to detect label flipping attacks and prevent them.	NSL-KDD, UNSW-NB15, Bank CreditCard, BRFSS, Diabetic	BYOL, KNN, SVM, DNN, FT-Transformer, and XGBoost	BYOL model performance had an F1 score of 77.9% in random label flipping, and 70.1% in decision boundary label flipping.	Should have included an extended framework of label flipping removal, sample reweighting, adaptive threshold, and robust retraining.

III. Methodology

As illustrated in Fig. 1, the proposed methodology consists of dataset preparation, baseline model training, simulation of label-flipping attacks, defense implementation, and performance evaluation.

A. Proposed Approach

First, we will prepare our dataset, the UNSW-NB15 dataset, which consists of benign traffic and nine types of network intrusion attacks (e.g., DoS, Exploits, Reconnaissance). The classification task will use the existing binary label provided in the dataset, where 0 represents (Benign) traffic, and 1 represents (Attack) traffic. We will also clean the dataset by removing missing or null values. We will then divide the dataset into training, validation, and test sets, following a 70%, 15%, and 15% split.

Next, we will train the ML models, specifically Random Forest and Decision Tree, on the clean training set and showcase the baseline NIDS performance before the attacks by testing them on

the test set. After that, we will poison the training set using a bidirectional label-flipping attack by changing the selected labels from attack (1) to benign (0), and vice versa, at poisoning rates of 10%, 20%, and 30% of the overall training set. The models will then be retrained on the poisoned set and evaluated on the clean test set. This will provide us with the opportunity to compare the performance of the system before and after the attacks and assess its vulnerability.

Next, we will apply a defense mechanism to help the system identify which instances may have flipped labels. For this, we will propose a hybrid approach combining KNN neighborhood label disagreement and out-of-fold model-label disagreement. If both defense mechanisms agree that the sample belongs to the opposite class, then it will be relabeled accordingly. For example, a suspicious benign sample might be relabeled as an attack, and a suspicious attack sample might be relabeled as benign. Otherwise, if the two mechanisms disagree or generate a low confidence score, the sample will instead be removed from the training set. The models will then be trained again on the cleaned set.

Finally, we will extensively evaluate and analyze where the defense works and where it fails, its computational cost, and its limitations. A further comparison will be made with recent papers to compare the performance of the different approaches.

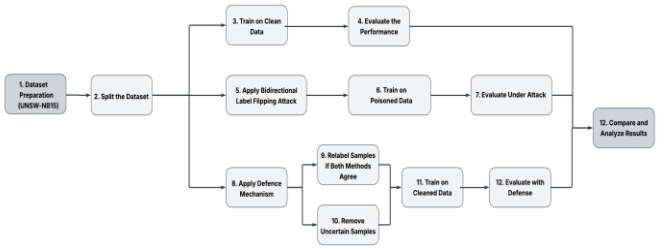


Fig. 1 Flowchart of the proposed methodology

B. Tools/Dataset/Technologies

Several tools and libraries will be used in our project to support preprocessing, coding, implementation, experimentation, and evaluation. One of the main tools is Google Colab, where we will primarily code in Python. Other tools include NumPy and Pandas, which are Python libraries that help perform mathematical operations and handle and clean the dataset. Scikit-learn, a machine-learning library, will also be used to train ML classifiers and calculates evaluation metrics. Additionally, Matplotlib will be used to create visualizations, such as graphs and charts, when needed. To support effective collaboration, GitHub will be used to store the project online, enabling us to edit the code and documentation and track members' contributions.

UNSW-NB15, a benchmark dataset containing network traffic classified as "Benign" and nine categories of malicious traffic, "Fuzzers", "Analysis", "Backdoors", "DoS" (Denial of Service), "Exploits", "Generic", "Reconnaissance", "Shellcode", and "Worms", will be the core dataset for system training and testing. The dataset is particularly useful for NIDS and contains a large number of network traffic samples, including their features and labels classifying them as either benign traffic or different types of attacks. For this project, its existing binary label will be used to classify network traffic.

As for the technologies our project will use, we will focus on machine learning and adversarial machine learning techniques for NIDS. This includes Random Forest and Decision Tree classifiers, as well as the simulation of bidirectional label-flipping data poisoning attacks.

C. System Design

1) Baseline Model:

Decision Tree and Random Forest are the baseline models for evaluating label-flipping attacks on a Network Intrusion Detection System (NIDS). As the dataset contains many network traffic features, Decision trees will learn nonlinear relationships between features to classify traffic into two classes: Benign (0) and Attack (1). Techniques such as pruning may be used to reduce overfitting when the tree becomes excessively complex.

In addition, Random Forest may improve generalization by combining multiple Decision Trees trained on random subsets of the data and features. Both models will first be trained and tested on the clean dataset. Their results, including accuracy, F1-score, and confusion matrix, will then be compared with those obtained from the poisoned

and defended models to evaluate the effect of the attack and the proposed defense method.

2) Attack Settings:

Label flipping will be applied to the training set at different poisoning ratios.

$$r \in \{0.10, 0.20, 0.30\}$$

A random Label-Flipping Attack (RLFA) will be used, where selected labels are randomly flipped according to the previously specified poisoning ratio. Since we're implementing a bidirectional attack, benign labels will be changed to attack, and attack labels will also be changed to benign. The poisoning ratio will be calculated based on the whole training dataset.

3) Defense Methods:

The proposed defense mechanism uses a hybrid method combining KNN neighborhood label disagreement and out-of-fold (OOF) model-label disagreement to identify suspicious labels. Before applying the defense, the original training labels will be stored separately as a ground truth and used only later to evaluate the effectiveness of the defense.

First, the KNN analysis will be used to identify suspicious samples by examining their nearest neighbors based on feature similarity. Since a label-flipping attack changes only the label without modifying the network traffic features, the poisoned sample will remain close to the other samples of its original class. For example, an attack sample that was poisoned and was incorrectly labeled as benign will still have its own attack features, making it surrounded by similar neighboring attack samples. This will therefore create some label inconsistency and neighborhood label disagreement.

The Out-Of-Fold method will then be applied for additional verification. The poisoned training dataset will be divided into multiple folds. In each iteration, the model will be trained on the remaining folds and used to predict the held-out fold. The OOF prediction will then be compared with the stored poisoned label. For example, if a sample was labelled as (benign) in the poisoned dataset but its KNN neighbors are (attack), and its OOF prediction is also (attack), the sample will be relabeled as (attack). Similarly, if a sample was labeled as (attack) but both KNN and OOF indicate (benign), it will be relabeled as benign. However, if KNN and OOF disagree, or if their confidence is low, the suspicious samples will instead be removed from the training set.

D. Evaluation Plan

The evaluation aims to measure how well the models classify network traffic and how effectively the proposed defense detects and corrects flipped labels.

First, models trained using the clean dataset, poisoned dataset, and defended dataset will be evaluated on the same clean test set. The performance of Random Forest and Decision Tree will be compared under these three conditions.

For the clean dataset, the evaluation establishes the baseline performance of the models. For the poisoned dataset, it measures the effect of the label-flipping attack on classification performance. Finally, for the defended dataset, the models are retrained using the corrected training data to determine how well the proposed defense mechanism restores the baseline classification performance. Metrics including accuracy, precision, recall, F1-score, confusion matrix, false-positive rate, and false-negative rate will be used to compare the models.

After that, a separate evaluation will be performed for the KNN + OOF defense mechanism. The suspicious samples will be compared with the original clean training labels stored before poisoning.

True Positive (TP) represents how many flipped labels were correctly detected; **True Negative (TN)** represents clean labels that were correctly left unchanged; **False Positive (FP)** represents how many clean labels were incorrectly identified as poisoned; and **False Negative (FN)** represents the number of flipped labels that were not detected. From these values, we can calculate defense precision, defense recall, defense F1-score, and clean retention rate.

E. Ethical and Legal Considerations

This research is undertaken with a sense of responsibility and is conducted for defensive and academic purposes and aims to improve the resilience of machine learning-based Network Intrusion Detection Systems (NIDS). Although label flipping techniques may be exploited to compromise machine learning-based security systems, this study

focuses on understanding their effects and proposing and evaluating mechanisms for mitigating them. Therefore, we provide information about the attack only to the extent that allows conducting evaluations of the hybrid KNN-OOF method.

In other words, all the label-flipping attacks will be simulated using copies of the UNSW-NB15 dataset in a controlled environment. No actual network, production system, or third-party device will be attacked or accessed without authorization. The dataset is used in accordance with its terms, meant for academic purposes only. No personal information or live network traffic will be collected as part of the experiments.

Following that, to preserve the integrity of the evaluation, we will ensure that the original clean labels are stored separately before poisoning, using them only as ground truth after applying the defense mechanism to evaluate its effectiveness in identifying poisoned samples and correcting them. The defense wouldn't have access to the original clean labels to prevent data leakage.

The possible impact of bidirectional label flipping will be evaluated using appropriate measures such as false negatives and false positives. Furthermore, clear experimental procedures and transparent performance metrics will be used to assess the proposed defence. The findings we will be reporting are meant for academic and defensive purposes, taking into account the possible exploitation of adversarial machine learning techniques.

IV. Team plan

This project is a collaborative effort by 4 members. We divided responsibilities equally among all members to ensure effective teamwork and plan the coming weeks to use our time as effectively as possible.

TABLE III
Task Breakdown

Team Member	Task Breakdown
Member 1- Reem Alomar	Defining the scope of our topic in NIDS, and the relevance of our problem to the real world where cybersecurity is needed. Stating the ethical and legal considerations we must take into account as we go into the effects of label-flipping, as well as the efficiency of our defenses.
Member 2- Jana Barrani	Stating our problem and covering our motivation for going through with label-flipping attacks in NIDS. Proposing a solution and defense to label-flipping attacks in NIDS setting, as well as the tools that will be used, datasets required, and technologies.
Member 3- Miriam Haider	Writing related work synthesis and what gaps were found accordingly in order to define the proposed approach that can be taken to resolve the problem of label-flipping attacks.
Member 4- Rand Gazzaz	Demonstrating the designed system based on the suggested approach of mitigating label-flipping attacks. Planning the evaluation metrics to be used to evaluate model performances in mitigating label-flipping attacks.

TABLE IIIII
Week 4-8 Timeline

Week	Expected Outcomes
Week 4	Finalizing and submitting Deliverable 1 through GitHub Repository
Week 5	Implementing and experimenting with our suggested approach to mitigate label flipping attacks. Starting with the dataset preprocessing and code implementation.
Week 6	Finishing code implementation, writing tests, and evaluating code. Fixing errors in implementation and comparing against baseline approaches.
Week 7	Documenting progress and updating report with limitations of approach, checking on ethical and legal compliance, and final architecture design.
Week 8	Finalizing the report and code implementation. Uploading it on GitHub Repository. Submitting Deliverable 2.

V. Conclusions

Adversarial attacks pose a significant challenge to the reliability and security of centralized machine-learning systems, as manipulating training data can affect model performance and decision-making. In Network Intrusion Detection Systems (NIDS), label-flipping attacks can modify labels between Benign and Attack, potentially causing malicious traffic to be classified as benign or legitimate traffic to be classified as malicious.

Existing research addresses two main aspects of this problem. Some studies, such as [2] and [5], investigate the impact of label-flipping attacks on model performance and demonstrate how increasing the proportion of flipped labels can degrade classification accuracy. Research addressing centralized systems has also investigated K-Nearest Neighbors (KNN) as a potential defense mechanism. Building on this research, the present study proposes a hybrid defense that combines KNN with an Out-of-Fold (OOF) prediction mechanism to identify potentially manipulated training labels. Disagreements between the observed labels and the predictions produced by these mechanisms are used as indicators of suspicious samples.

The evaluation focuses on two complementary aspects using clean, poisoned, and defended training datasets, as well as evaluating the effectiveness of the defense in detecting flipped labels.

Overall, the study aims to investigate the potential of combining KNN and OOF predictions as a defense against label-flipping attacks in a centralized NIDS and to identify the strengths and limitations of the proposed approach.

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